

choice allows us to relax the parallel trends assumption to be just $E[Y_{i,e+\ell}^\infty - Y_{i,0}^\infty | E_i = e] = E[Y_{i,e+\ell}^\infty - Y_{i,0}^\infty | E_i > e + \ell]$. If instead we want to relax Assumption 2 to allow some anticipation, we then need the parallel trends assumption to hold. For example, suppose we are only willing to assume away anticipation 2 periods before a unit is treated, we may still be able to recover several $CATT_{e,\ell}$ by appropriately selecting a pre-period and a smaller set of control cohorts. We may choose any $s < e - 2$ for the pre-period, and for control cohorts, we may choose cohorts treated only after at least 2 periods from now $C \subseteq \{c : e + \ell + 2 < c \leq T\}$, or choose the latest-treated cohort $C = \{\max\{E_i\}\}$ as specified in regression (26).

5 Empirical Illustration

We illustrate our findings in the setting of Dobkin et al. (2018). Dobkin et al. (2018) study the economic consequences of hospitalization, which is a large source of economic risk for adults in the United States. To quantify these economic risks, in the first part of their analysis, Dobkin et al. (2018) leverage variation in the timing of hospitalization observed in the publicly-available dataset, Health and Retirement Study (HRS), which we describe in more detail in Section 5.1. Their estimation of the dynamic effects of hospitalization using two-way fixed effects regressions provides a good context for demonstrating our results. As we argue below in Section 5.2, parallel trends and no anticipation assumptions are plausible in this setting. However, the effect of hospitalization is potentially heterogenous across individuals hospitalized in different years. Our findings of non-convex and non-zero weighting would therefore apply to their two-way fixed effects regression estimates, and our alternative estimator could lead to different estimates. Furthermore, this dataset is publicly available, which allows us to provide replication files.

5.1 Data

Our sample selection closely follows Dobkin et al. (2018) but we include a cursory explanation here for completeness with an emphasis on how our final sample differs from their main analysis sample. Our primary source of data is the biennial Health and Retirement Study (HRS). We identify the sample of individuals who appear in two sequential waves of surveys and newly report having a hospital admission over the last two years (the “index” or initial admission) at the second survey. To focus on health “shocks”, we restrict attention to non-pregnancy-related hospital admissions as in Dobkin et al. (2018). We also follow Dobkin et al. (2018) by focusing on adults who are hospitalized at ages 50-59.

Unlike Dobkin et al. (2018), we restrict our analysis to a subsample of these individuals who appear throughout waves 7-11 (roughly 2004-2012). Our sample of analysis therefore includes HRS respondents with index hospitalization during waves 8-11. The purpose of this sample restriction is to maintain a balanced panel with a reasonable sample size.

Here i indexes an individual, and t indexes survey wave ($T = 4$) and is normalized to zero for wave 7, the first wave in our sample. Among the outcomes $Y_{i,t}$ studied by Dobkin et al. (2018), we focus on two: out-of-pocket medical spending and labor earnings. They are derived from self-reports, adjusted to 2005 dollars and censored at the 99.95th percentile.

Summary statistics. Table 2 presents basic summary statistics for our analysis sample before hospitalization. We have a slightly lower fraction of white in our sample, but otherwise have a similar sample to Dobkin et al. (2018).

In Panel D, we compare means of the cross-sectional distributions of outcomes for individuals who have not been hospitalized by each wave. The size of the sample conditional on not having been hospitalized strictly decreases with each subsequent wave. There are apparent time trends in our outcomes of interest prior to hospitalization as we observe distributional changes across waves. Out-of-pocket medical spendings fluctuate and earnings decrease with each wave on average as more individuals are retired in each subsequent wave.

5.2 Setting

We illustrate how variations in the timing of hospitalization fit the event studies design proposed in Section 2. We define treatment $D_{i,t}$ to be ever having been hospitalized. In our terminology, we categorize individuals into cohorts based on E_i , which is defined as the survey wave of their initial hospitalization. Since we restrict the sample to individuals who were ever hospitalized in waves 8-11, there are four cohorts $E_i \in \{1, 2, 3, 4\}$. Although hospitalization itself may not be an absorbing state, we are trying to model the impact of having had any hospitalization. Thus, the cohort-specific average treatment effects $CATT_{e,\ell}$ trace out the path of treatment effects for cohort e following a negative health shock (even though the shock itself may be transient), as opposed to never having been hospitalized. Next we discuss whether each of the three identifying assumptions proposed is likely to hold in the context of unexpected hospitalizations.

Parallel trends (Assumption 1). Hospitalization is likely to be earlier among sicker individuals with high out-of-pocket medical spending and low labor earnings. Thus, it is not plausible that the baseline

outcome $Y_{i,t}^\infty$ is mean independent of the timing of hospitalization. The parallel trends assumption is more plausible as it allows the timing to depend on unobserved time-invariant characteristics such as chronic disease. Furthermore, hospitalized individuals might be on a downward trend for labor earnings already prior to hospitalization compared to individuals who are never hospitalized. To reduce such confounding, we follow Dobkin et al. (2018) to restrict the parallel trends assumption to individuals who were ever hospitalized.

No anticipatory behavior (Assumption 2). It is plausible that there is no anticipatory behavior prior to the hospitalization, given that the treatment is restricted to conditions that are likely unexpected hospitalizations. This assumption may be violated if individuals have private information about the probability of these hospitalizations over time and thus adjust their behavior prior to hospitalization.

Treatment effect heterogeneity (Assumptions 3). For out-of-pocket medical spending, the effect of hospitalization is potentially heterogenous across individuals hospitalized in different waves. Individuals hospitalized in later waves are mechanically older at the time of hospitalization than individuals hospitalized in earlier waves. The effect on out-of-pocket medical spending is largely determined by generosity of health insurance, which may decrease as individuals age into Medicare. The effect on labor earnings is also likely heterogenous as it depends on the labor market condition at the time of hospitalization: for example, individuals hospitalized during the financial crisis may find it more difficult to return to the labor force, and suffer a more grave decrease in earnings.

5.3 Illustrating weights in two-way fixed effects regression

We illustrate our results on two-way fixed effects regression by estimating the following specification with indicators for up to three leads and lags following equation (3) of Dobkin et al. (2018)

$$Y_{i,t} = \alpha_i + \lambda_t + \mu_{-3}D_{i,t}^{-3} + \mu_{-2}D_{i,t}^{-2} + \mu_0D_{i,t}^0 + \mu_1D_{i,t}^1 + \mu_2D_{i,t}^2 + \mu_3D_{i,t}^3 + v_{i,t}. \quad (29)$$

For their estimation, Dobkin et al. (2018) trim their sample, keeping only observations up to three waves prior to the hospitalization and three waves after the hospitalization, and weight their regression with survey weights. To fully illustrate issues with this specification, we do not trim, but rather use a sample balanced in calendar time for $t \in \{0, \dots, 4\}$, and do not apply survey weights. With a sample balanced in calendar time, we need to exclude at least two relative period indicators due to multicollinearity. Following Dobkin et al. (2018), we exclude the period right before hospitalization ($\ell = -1$). We also exclude $\ell = -4$. Since we do

not trim, note that our results are not directly comparable to Dobkin et al. (2018) even though they are quite similar.

We focus on a single coefficient μ_{-2} that is supposed to test for any pre-trend of hospitalizations. As in Proposition 2, we can decompose μ_{-2} as

$$\sum_{e=1}^4 \omega_{e,0}^{-2} CATT_{e,0} + \sum_{\ell=-3, \neq -1}^3 \sum_{e=1}^4 \omega_{e,\ell}^{-2} CATT_{e,\ell} + \sum_{\ell'=-4, -1} \sum_{e=1}^4 \omega_{e,\ell'}^{-2} CATT_{e,\ell'}. \quad (30)$$

As discussed in Proposition 1 we can estimate the underlying weights $\omega_{e,\ell}^{-2}$ by regressing $\mathbf{1}\{E_i = e\} \cdot D_{i,t}^\ell$ on the relative wave indicators included in specification (29) i.e. $\{D_{i,t}^\ell\}_{\ell=-3, \neq -1}^3$ and two-way fixed effects. The coefficient estimator of $D_{i,t}^{-2}$ in such regression, $\widehat{\omega}_{e,\ell}^{-2}$, consistently estimates $\omega_{e,\ell}^{-2}$.

Figure 3 plots these estimated weights. As described in our decomposition results, these weights have the following properties: (a) the four weights from relative wave $\ell = -2$ sum to one; (b) the weights for other included relative waves $\ell \in \{-3, 0, 1, 2, 3\}$ sum to zero for each included relative wave; and (c) the weights from excluded relative waves $\ell \in \{-4, -1\}$ sum to negative one across these excluded relative waves.

The weights are non-negative for lags of treatments, which suggest that the FE estimate $\widehat{\mu}_{-2}$ is particularly sensitive to estimates of the dynamic effects of hospitalizations and does not isolate the pre-trends. Specifically, treatment effects heterogeneity in $\ell = -3, 0, 1, 2$ can affect the FE estimate $\widehat{\mu}_{-2}$. Applied researchers can make similar plots to visualize the role of weights in their settings with our publicly-available Stata package `eventstudyweights` by Sun (2020).

5.4 Comparing FE and IW estimates

We illustrate our alternative method (IW estimator) following the three steps outline in Section 4.1. First, we estimate the interacted specification (26) as

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{e \in \{1, 2, 3\}} \sum_{\ell=-3, \neq -1}^2 \delta_{e,\ell} \mathbf{1}\{E_i = e\} \cdot D_{i,t}^\ell + \epsilon_{i,t} \quad (31)$$

for $t = 0, \dots, 3$. Specifically, we estimate $CATT_{e,\ell}$ using a DID estimator $\widehat{\delta}_{e,\ell}$ with pre-period $s = -1$ and control cohort $C = 4$, the cohort hospitalized in the last period. This means we need to drop $t = 4$ from estimation because everyone has been hospitalized by $t = 4$, and a control cohort for estimating $CATT_{e,\ell}$ in $t = 4$ does not exist. Second, we estimate the sample share of each cohort e across cohorts that experience